

Tomás P. Lenzi

M.Sc. student and Researcher

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Summary

- Master's student in Computer Science at PUC-Rio and full-time researcher at TecGraf, with research interests in generative artificial intelligence, natural language processing, and natural-language interfaces to structured data. I completed my B.Sc. in Computer Science at PUC-Rio in July 2026, developing an undergraduate thesis on the systematic evaluation of conversational text-to-SQL agents, with results published in co-authored papers. At TecGraf I build production language-model systems for large corporate clients. In the Master's I am deepening evaluation methods and information retrieval over structured databases mediated by LLMs.

Education

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| M.Sc. Pontifical Catholic University of Rio de Janeiro (PUC-Rio), Computer Science | Rio de Janeiro, Brazil
Aug 2026 – present |
| B.Sc. Pontifical Catholic University of Rio de Janeiro (PUC-Rio), Computer Science | Rio de Janeiro, Brazil
Aug 2022 – July 2026 |
- Teaching Assistant, Applied Computing (Mar 2025 – Jul 2025).

Experience

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| TecGraf, Pontifical Catholic University of Rio de Janeiro (PUC-Rio), Researcher | Rio de Janeiro, Brazil
Sept 2025 – present
1 year |
| PVX Labs, Software Developer Intern | Remote
Jan 2025 – July 2025
7 months |
| Pacifica Continental, Business Intelligence Intern | Rio de Janeiro, Brazil
Mar 2022 – Dec 2023
1 year 10 months |
- Conducted research on evaluating LLM-based conversational agents, proposing new strategies for automatic test-case generation and running comparative experiments across multiple benchmarks, with results published in co-authored papers.
 - Investigating the integration of LLM-based agents into a large energy-sector client's legacy systems, evaluating orchestration architectures and production deployment viability.
 - Design and implement application architectures for production LLM agents, including FastAPI services, LangGraph orchestration, and SSE streaming.
 - Contributed to the development and maintenance of open-source libraries for machine learning in Elixir.
 - Conducted formal comparisons between Elixir and Python, evaluating performance, usability, and suitability for ML applications.
 - Conducted statistical studies and implemented optimization and scalability processes for commercial activities across Brazil, Latin America, the United States, and Europe.

Volunteer

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| Pré Vestibular Comunitário da PUC-Rio, Mathematics Teacher
Community college-prep program at PUC-Rio. | Rio de Janeiro, Brazil
Mar 2024 – present |
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- Teach high-school mathematics with a focus on preparation for the ENEM, UERJ, and PUC-Rio entrance exams.
 - Design questions and materials for mock exams aligned with Brazil's main university entrance exams.

Publications

Benchmarking natural language database conversational agents

2026

T. P. Lenzi, E. R. S. Nascimento, M. O. Silva, G. M. García, Y. T. Izquierdo, M. S. P. Facina, I. G. Siqueira, M. Lemos, M. A. Casanova
[10.1007/978-3-032-28160-9_1](https://doi.org/10.1007/978-3-032-28160-9_1) (Advanced Information Systems Engineering Workshops – CAiSE 2026 (LNBIP 586). Springer)

Automated evaluation of conversational text-to-SQL agents by schema traversal

2026

T. P. Lenzi, E. R. S. Nascimento, M. O. Silva, G. M. García, Y. T. Izquierdo, I. G. Siqueira, A. C. Finamore, M. Lemos, M. A. Casanova
(SN Computer Science (manuscript submitted for publication))

Skills

Artificial Intelligence: Large Language Models, Generative AI, Natural Language Processing, Text-to-SQL, Conversational Agents, Agent Evaluation

Software Engineering: Python, FastAPI, LangGraph, SSE, Elixir

Data: Business Intelligence, Statistical Analysis

Languages

Portuguese: Native speaker

English: Fluent

Interests

Research

- Generative AI
- Natural Language Processing
- Natural-language interfaces to structured data
- Evaluation of conversational agents
- Information retrieval over structured databases

Certificates

Angular Framework

Data Science